

Week 4: Algebra in Context

Jacob Khohayting

Last updated: September 20, 2026

Contents

Word Problems	2
Unit Conversions	3
Rate Problems	4
Practice Problems	7

※ Word Problems

To your surprise, math isn't just numbers and equations. In these problems, you will have to decode what a problem is saying. We saw some of these problems last week, but this week we will go further into analyzing them. Then, we will dive into the infamous "rate problems".

Two baskets contain 26 apples altogether. The red basket contains 4 more apples than the blue basket. How many apples are in each basket?

Let x be the number of apples in the blue basket. The red basket has 4 more, so it contains $x + 4$ apples. Since there are 26 apples altogether,

$$x + (x + 4) = 26.$$

Solving,

$$2x + 4 = 26 \implies 2x = 22 \implies x = 11.$$

Therefore, the blue basket contains 11 apples and the red basket contains $11 + 4 = \boxed{15}$ apples. Checking, $11 + 15 = 26$ and $15 - 11 = 4$.

This problem could also be written as a system:

$$\begin{cases} b + r = 26, \text{ and} \\ r = b + 4, \end{cases}$$

where r and b denote the number of apples in red and blue baskets, respectively.

In general, a variable should stand for a particular quantity. Include the unit when you define it. For example, on your test papers, feel free to write

$$x = \text{the number of books on the first shelf,}$$

not just "let x be the answer." A useful process is:

1. **Find the question.** What quantity must you discover?
2. **Name the unknown.** Let a letter stand for it.
3. **Describe the other quantities.** Write them using the same variable when possible.
4. **Write an equation.** Every number in the equation should have a job in the story.
5. **Solve and check.** Put the answer back into the story.

In general, there are many types of "relationships" that you should know:

Story phrase	Algebra
7 more than x	$x + 7$
7 less than x	$x - 7$
twice x	$2x$
three times x	$3x$
half of x	$\frac{x}{2}$
x shared equally among 4 people	$\frac{x}{4}$

Keywords are clues, but they are not enough by themselves. Read the entire sentence. For example, “5 less than x ” means $x - 5$, not $5 - x$.

Frankly, I won’t dwell on this too much since this was covered last week. There is a large amount of practice in the practice problems, however.

✧ Unit Conversions

Which is longer: 5 feet or 58 inches? By how much?

Since a foot consists of 12 inches, we have 5 feet = $5 \cdot 12 = 60$ inches. Now both measurements use the same unit: 60 inches > 58 inches. Therefore, 5 feet is longer by $60 - 58 = \boxed{2}$ inches.

Essentially, what we did here is convert units. A unit conversion changes the *name* of a measurement, but it does not change its actual size. For example,

$$1 \text{ foot} = 12 \text{ inches.}$$

These are two names for the same length. A conversion fact is also a special kind of rate: 12 inches per foot. Here is the method:

1. Decide which unit you want.
2. Write a conversion fact connecting the two units.
3. Multiply or divide so that the unwanted unit disappears.

For example, for the above example,

$$5 \text{ feet} \times \frac{12 \text{ inches}}{1 \text{ foot}} = 60 \text{ inches.}$$

It is crucial that you understand this is valid since the fraction we are multiplying by here is basically 1. The numerator and denominator are the same. Consider another example:

Four notebooks cost \$12. How much do 7 notebooks cost?

We have this unit conversion:

$$\frac{12 \text{ dollars}}{4 \text{ notebooks}}.$$

This fraction is basically 1 since the numerator and denominator are the same thing. And so,

$$7 \text{ notebooks} = 7 \text{ notebooks} \times \frac{12 \text{ dollars}}{4 \text{ notebooks}} = 21 \text{ dollars,}$$

as the notebooks cancel out. Here is a list of important facts you should know:

Type	Conversion facts
Length	1 foot = 12 inches 1 yard = 3 feet = 36 inches
Time	1 minute = 60 seconds 1 hour = 60 minutes
Weight	1 pound = 16 ounces
Money	1 dollar = 100 cents
Metric length	1 meter = 100 centimeters 1 kilometer = 1000 meters
Metric mass	1 kilogram = 1000 grams
Metric capacity	1 liter = 1000 milliliters

A robot moves at 12 feet per second. How many yards does it travel in 5 seconds?

First find the distance in feet:

$$5 \text{ seconds} = 5 \text{ seconds} \cdot \frac{12 \text{ feet}}{1 \text{ second}} = 60 \text{ feet.}$$

Then convert feet to yards:

$$60 \div 3 = \boxed{20 \text{ yards}}.$$

A 3-meter rope is cut into 6-centimeter pieces. How many pieces are made?

We will convert all the units to the same value to get a better hold. We know that a meter is equivalent to 100 centimeters. So,

$$3 \text{ meters} \times \frac{100 \text{ centimeters}}{1 \text{ meters}} = 300 \text{ centimeters.}$$

The number of pieces made is $\frac{300}{6} = 50$.

✂ Rate Problems

A toy car travels 6 feet in 2 seconds. It continues at the same speed. How far will it travel in 5 seconds?

First, we can create the unit conversion

$$\frac{6 \text{ feet}}{2 \text{ seconds}}$$

to compare feet and seconds. Thus,

$$5 \text{ seconds} = 5 \text{ seconds} \times \frac{6 \text{ feet}}{2 \text{ seconds}} = \boxed{15 \text{ feet}}.$$

In general, a **rate** compares two quantities that have different units. Examples include

3 feet per second,
10 miles per hour,
4 pages per minute,
5 dollars per notebook.

The word *per* means “for each.” For example,

3 feet per second

means 3 feet for each 1 second. A rate that compares a quantity to one unit is called a **unit rate**. In most of the problems in this section, the rate is constant. A **constant rate** means that the same amount is traveled, made, or used during each unit of time.

For motion at a constant speed,

$$\text{distance} = \text{rate} \times \text{time}.$$

Using letters, we write $d = rt$ where d denote distance, r is rate, and t is time. We can rearrange the equation depending on what we need to find:

$$r = \frac{d}{t} \quad \text{and} \quad t = \frac{d}{r}.$$

The time unit must match the time unit in the rate. For example, a rate of 10 miles per hour must be multiplied by a number of hours:

$$\left(10 \frac{\text{miles}}{\text{hour}}\right) \times (2 \text{ hours}) = 20 \text{ miles}.$$

The hours cancel, leaving miles. When a rate is given in miles per hour but the time is given in minutes, convert the minutes to hours:

$$20 \text{ minutes} = \frac{20}{60} \text{ hour} = \frac{1}{3} \text{ hour},$$

When an answer is found in hours but the problem asks for minutes, multiply by 60:

$$\frac{1}{10} \text{ hour} = \frac{1}{10} \cdot 60 = 6 \text{ minutes}.$$

Jack and Jill are going swimming at a pool that is one mile from their house. They leave home simultaneously. Jill rides her bicycle to the pool at a constant speed of 10 miles per hour. Jack walks to the pool at a constant speed of 4 miles per hour. How many minutes before Jack does Jill arrive?

Both Jack and Jill travel 1 mile. We can use the previous $d = rt$ formula. Jill's travel time is

$$t = \frac{d}{r} = \frac{1 \text{ mile}}{10 \text{ miles / hour}} = \frac{1}{10} \text{ hour} = \frac{1}{10} \text{ hour} \times \frac{60 \text{ minutes}}{1 \text{ hour}} = 6 \text{ minutes}.$$

So, Jill takes 6 minutes. Now, for Jack's travel time is

$$t = \frac{d}{r} = \frac{1 \text{ mile}}{4 \text{ miles / hour}} = \frac{1}{10} \text{ hour} = \frac{1}{4} \text{ hour} \times \frac{60 \text{ minutes}}{1 \text{ hour}} = 15 \text{ minutes.}$$

So, Jack takes 15 minutes. Therefore, Jill arrives $15 - 6 = \boxed{9}$ minutes before Jack.

Jeremy's father drives him to school in rush-hour traffic in 20 minutes. One day there is no traffic, so his father can drive 18 miles per hour faster and gets him to school in 12 minutes. How far is it to school?

Let r denote the rush-hour speed in miles per hour. Then, the no-traffic speed is $r + 18$. Convert both travel times to hours:

$$20 \text{ minutes} = \frac{1}{3} \text{ hour} \quad \text{and} \quad 12 \text{ minutes} = \frac{1}{5} \text{ hour.}$$

The distance to school is the same on both days. In rush-hour traffic, the distance is

$$\underline{d = rt} = r \left(\frac{1}{3} \right) = \frac{r}{3}.$$

With no traffic, the distance is

$$\underline{d = rt} = (r + 18) \left(\frac{1}{5} \right) = \frac{r + 18}{5}.$$

Setting the two distance expressions equal and solving:

$$\frac{r}{3} = \frac{r + 18}{5} \implies 5r = 3(r + 18) = 3r + 54 \implies r = 27.$$

The rush-hour speed is 27 miles per hour. Therefore, the distance to school is

$$d = rt = 27 \left(\frac{1}{3} \right) = \boxed{9 \text{ miles}}.$$

And finally, average speed is not always found by adding two speeds and dividing by two. The correct formula is

$$\text{average speed} = \frac{\text{total distance}}{\text{total time}}.$$

For a trip with several parts:

1. Find the distance and time for each part.
2. Add all the distances.
3. Add all the times.
4. Divide the total distance by the total time.

If a traveler spends more time moving at a slower speed, then because of this, the average of two speeds may not be the average speed for the whole trip.

※ Practice Problems

1. Anna and Bella are celebrating their birthdays together. Five years ago, when Bella turned 6 years old, she received a newborn kitten as a birthday present. Today the sum of the ages of the two children and the kitten is 30 years. How many years older than Bella is Anna?
2. Luka is making lemonade to sell at a school fundraiser. His recipe requires 4 times as much water as sugar and twice as much sugar as lemon juice. He uses 3 cups of lemon juice. How many cups of water does he need?
3. Abby, Bob, and Cathy each have some marbles. Abby has 36 and Bob has 22. Abby gives half her marbles to Bob, who gives half his combined marbles to Cathy. In the end, Cathy has 29 marbles. How many marbles did Cathy start with?
4. Bella begins to walk from her house toward her friend Ella's house. At the same time, Ella begins to ride her bicycle toward Bella's house. They each maintain a constant speed, and Ella rides 5 times as fast as Bella walks. The distance between their houses is 2 miles, which is 10,560 feet, and Bella covers $2\frac{1}{2}$ feet with each step. How many steps will Bella take by the time she meets Ella?
5. In a middle-school mentoring program, a number of the sixth graders are paired with a ninth-grade student as a buddy. No ninth grader is assigned more than one sixth-grade buddy. If $\frac{1}{3}$ of all the ninth graders are paired with $\frac{2}{5}$ of all the sixth graders, what fraction of the total number of sixth and ninth graders have a buddy?
6. Bridget bought a bag of apples at the grocery store. She gave half of the apples to Ann. Then she gave Cassie 3 apples, keeping 4 apples for herself. How many apples did Bridget buy?
7. Rohan keeps a total of 90 guppies in 4 fish tanks.
 - There is 1 more guppy in the 2nd tank than the 1st tank.
 - There are 2 more guppies the the 3rd tank than the 2nd tank.
 - There are 3 more guppies in the 4th tank than the 3rd tank.

How many guppies are in the 4th tank?

8. When the World Wide Web first became popular in the 1990s, download speeds reached a maximum of about 56 kilobits per second. Approximately how many minutes would the download of a 4.2-megabyte song have taken at that speed? (Note that there are 8000 kilobits in a megabyte.)
9. Two towns, A and B , are connected by a straight road, 15 miles long. Traveling from town A to town B , the speed limit changes every 5 miles: from 25 to 40 to 20 miles per hour (mph). Two cars, one at town A and one at town B , start moving toward each other at the same time. They drive exactly the speed limit in each portion of the road. How far from town A , in miles, will the two cars meet?
10. A driver travels for 2 hours at 60 miles per hour, during which her car gets 30 miles per gallon of gasoline. She is paid \$0.50 per mile, and her only expense is gasoline at \$2.00 per gallon. What is her net rate of pay, in dollars per hour, after this expense?
11. A box contains 10 pounds of a nut mix that is 50 percent peanuts, 20 percent cashews, and 30 percent almonds. A second nut mix containing 20 percent peanuts, 40 percent cashews, and 40 percent

- almonds is added to the box resulting in a new nut mix that is 40 percent peanuts. How many pounds of cashews are now in the box?
12. The weight of $\frac{1}{3}$ of a large pizza together with $3\frac{1}{2}$ cups of orange slices is the same as the weight of $\frac{3}{4}$ of a large pizza together with $\frac{1}{2}$ cup of orange slices. A cup of orange slices weighs $\frac{1}{4}$ of a pound. What is the weight, in pounds, of a large pizza?
 13. Mr. Lopez has a choice of two routes to get to work. Route A is 6 miles long, and his average speed along this route is 30 miles per hour. Route B is 5 miles long, and his average speed along this route is 40 miles per hour, except for a $\frac{1}{2}$ -mile stretch in a school zone where his average speed is 20 miles per hour. By how many minutes is Route B quicker than Route A?
 14. At a math contest, 57 students are wearing blue shirts, and another 75 students are wearing yellow shirts. The 132 students are assigned into 66 pairs. In exactly 23 of these pairs, both students are wearing blue shirts. In how many pairs are both students wearing yellow shirts?
 15. Mrs. Jones is pouring orange juice for her 4 kids into 4 identical glasses. She fills the first 3 full, but only has enough orange juice to fill one third of the last glass. What fraction of a glass of orange juice does she need to pour from the 3 full glasses into the last glass so that all glasses have an equal amount of orange juice?
 16. Cities A and B are 45 miles apart. Alicia lives in A and Beth lives in B . Alicia bikes towards B at 18 miles per hour. Leaving at the same time, Beth bikes toward A at 12 miles per hour. How many miles from City A will they be when they meet?